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Are you hallucinated? Insights into large language models

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ABSTRACT

The work presented in the paper challenges the common view of hallucination as a flaw in Large Language Models (LLMs), proposing instead that it is an inherent feature of their architecture. It explores the relationship between "creativity," "generalization," and "hallucination," showing that they result from the transformer model's high-dimensional vector blending. Using Gemini 2.0 and 2.5 models, the authors demonstrate a "Self-Correction Blind Spot," where LLMs can recognize errors but lack the reasoning pathways to correct them. The paper introduces the "Generalization-Hallucination Trade-off," suggesting that minimizing hallucination would impede generalization. The research advocates for managing hallucination with external verification rather than eliminating it.

1. Introduction

The rapid expansion of Large Language Models (LLMs) into diverse sectors has been met with both enthusiasm for their capabilities and significant concern regarding their accuracy and reliability [1]. A primary concern is *hallucination*, where models generate outputs that are seemingly plausible and logical yet factually incorrect or unfaithful to the source [1]. This tendency is viewed as a critical flaw of LLMs, which hinders their progress in several high-stakes domains, including law and medicine [2–6].

1.1. Defining and categorizing hallucinations

The academic definition of hallucination has a clear distinction between a failure of *factualness* (misalignment with world knowledge) and a failure of *faithfulness* (divergence from a given source) [1]. This separation is critical and is detailed in Table 1. It is also further classified as *entity-error* (wrong name) and *relation-error* (wrong relationship) [7].

An LLM is a probabilistic system that lacks consciousness and a concept of reality; it just predicts the next likely token. An incorrect output is not an error of code but a value judgment assigned by us. The model is functioning exactly as designed, not deviating from a truth-seeking baseline that never existed [8,9].

1.2. Related work and knowledge gaps

Broad reviews of natural language processing [5] and specific studies on LLM capabilities [6] highlight the exponential growth of these models across diverse domains, from mental health journaling [4] to sign language recognition.

However, a significant knowledge gap persists. Most current research treats hallucination strictly as a harmful error to be eradicated [10]. For instance, in high-stakes medical or legal applications, hallucinations are viewed purely as reliability failures [2]. While valid, this "defect-centric" view often overlooks the architectural necessity of the phenomenon. This research fills this gaps by analyzing hallucination not just as a failure of *factualness*, but as a functional consequence of the vector association mechanisms — such as those found in token-selective transformers [11] — that drive the model's generative power. Multimodal selective-attention frameworks such as SoVTP further describe how token- and region-level filtering influences hallucination dynamics in cross-modal models [12].

1.3. Motivation and contribution

The paper challenges the conventional view of hallucination in LLMs, reframing it as an inherent feature rather than a flaw. Traditionally, hallucination is considered a critical error of LLMs to be eliminated, particularly in high-stakes domains like medicine and law.

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Table 1
Taxonomy of hallucination in LLMs.

Category	Sub-type	Definition and example
Factualness	Factual fabrication	Generation of unverifiable information. *e.g., citing a non-existent academic paper.*
	Factual contradiction	Generation of information that contradicts established world knowledge. *e.g., stating the Eiffel Tower is in Berlin.*
Faithfulness	Intrinsic hallucination	Output contradicts or deforms information within a provided source. *e.g., reporting a $10M profit as $100M when summarizing a source.*
	Extrinsic hallucination	Output contains information not verifiable from the source. *e.g., adding general background info to a paper summary.*

However, it is argued that hallucination is a necessary outcome of the LLM's architecture; it arises from its ability to extrapolate from high-dimensional latent spaces. This analysis is essential because it helps us understand the balance between creativity and factual accuracy, which is central to the effective application of LLMs. By examining hallucination, we can explore its potential in domains where creative reasoning and generalization are needed, while managing its risks in contexts requiring precision. The analysis introduces the *Generalization–Hallucination Trade-off*, demonstrating that minimizing hallucination limits the model's ability to generalize and handle novel situations. It also identifies the *self-correction blind spot*, where LLMs can detect errors but lack the pathways to correct them autonomously. Another novelty is the analysis of hallucination, driven by the high-dimensional vector blending in transformers, which is a vital mechanism for reasoning and creative generation. This reframing of hallucination as an essential feature shifts the focus from eliminating it to managing it strategically, depending on the application context. Furthermore, through a controlled study of Gemini 2.0 and 2.5, we provide a novel *deterministic evidence* that under zero-temperature conditions ($T = 0$), hallucination is a reproducible artifact of latent geometry rather than random perturbation, offering a unified perspective that formally links extrapolation, the generalization–hallucination balance, and the self-correction limitation. The paper highlights how this new perspective on hallucination opens the door for better utilization and control of LLMs in various domains.

Organization of the paper

The rest of the paper is organized as follows: Section 2 enlightens the mathematical model of LLM architecture. Section 3 discusses a model for hallucination. Section 4 discusses hallucination and approximation. Section 5 highlights the analysis of self-correction limitations. Section 6 focuses on empirical and methodological constraints, while Section 7 illustrates contemporary practices for mitigation, evaluation, and application. Finally, Section 8 concludes the paper with future scope.

2. Mathematical model of LLM architecture

An LLM is a deterministic function implementing auto-regressive prediction via the Transformer architecture [13]. It processes token embeddings through multi-head self-attention and feed-forward networks to navigate a geometric latent space.

2.1. Prediction and latent space geometry

The causal objective is to predict the next token based on context [14], expressed via the chain rule:

$$P(w_1, \dots, w_n) = \prod_{i=1}^{n} P(w_i | w_1, \dots, w_{i-1}) \tag{1}$$

Operations occur in a high-dimensional space where semantic proximity holds [9,15]. Analogical reasoning emerges as consistent vector offsets:

$$\vec{v}_{\text{king}} - \vec{v}_{\text{man}} + \vec{v}_{\text{woman}} \approx \vec{v}_{\text{queen}} \tag{2}$$

This mathematical structure effectively mirrors cognitive organization [16–18].

2.2. Determinism and its practical limits

While a temperature of 0 theoretically guarantees determinism, perfect reproducibility is constrained by hardware and architectural realities [19]:

1. *Floating-Point Associativity:* GPU arithmetic is not strictly associative ($(a + b) + c \neq a + (b + c)$). Parallel summation order variability causes minute rounding differences that alter token ranking [20].
2. *MoE Routing:* In Mixture-of-Experts models, token routing depends on batch composition and concurrent load, altering the computational path [19].

Despite these nuances, the underlying mathematical mapping remains deterministic.

3. A model for hallucination

Hallucinations are the logical consequence of the transformer's essential mathematical operation: the self-attention mechanism. This mechanism allows the model to weigh the importance of different tokens in the input sequence when computing an expression for the next token. Its mathematical formulation is given by [13]:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3}$$

where Q (Queries), K (Keys), and V (Values) are matrices derived from the input token embeddings, and d_k is the dimension of the key vectors. The softmax$(\cdot)$ function computes a matrix of attention weights based on query-key similarity [13].

Fundamentally, this operation is a refined form of weighted averaging. Each output vector is a weighted sum of the value (V) vectors. This mathematical merging is precisely what enables the model to create a new vector representation in the latent space that does not correspond to any single input token but is a synthesis of multiple concepts [9] . This is where the *curse of dimensionality* becomes vital. The latent space's volume increases exponentially with the number of dimensions, making the limited training data points a lot more sparse. Consequently, for any novel query, the resulting vector is almost certain to fall into a vast, *empty* region of the space, far from any particular training examples. This makes extrapolation the default mode of operation, not an exception. The probability of a new point falling neatly

Table 2
Comparison of traditional database vs. large language model.

Attribute	Traditional Database	Large Language Model (LLM)
Primary Function	Deterministic data storage and retrieval.	Probabilistic pattern recognition and text generation.
Data Storage	Explicit, structured, discrete records.	Implicit, distributed statistical patterns in model weights.
Response to Novelty	Returns "no result found" or error.	Generates a probabilistic, *best-guess* answer via extrapolation.
Output Nature	Deterministic and retrieved.	Generative and probabilistic.

within the convex hull of training points approaches zero, meaning generalization is almost always *an act of extrapolation* [19].

A hallucination occurs when the model decodes one of these extrapolated vectors. The result is a very reasonable but ungrounded completion, in other words, the model's best guess at what should exist in an unexplored region of its conceptual map. For instance, our brain compensates for the physiological blind spot in the retina (caused by the absence of photoreceptor at the optic nerve) by interpolating patterns from adjacent regions, automatically filling in missing sensory input to maintain perceptual coherence [7]. Similarly, the cognitive discovery of object permanence in infants demonstrates an innate predictive faculty, wherein the mind infers the continued existence of unseen objects based on prior exposure—a primitive form of auto-completion [21]. In both cases, humans and machines alike rely on probabilistic inference to extend beyond available information, underscoring hallucination not as a malfunction but as a natural byproduct of predictive intelligence.

A foundational characteristic of human language is its reliance on generalization and inference, operating upon an underlying layer of shared context rather than clearly descriptive data alone [22]. For an artificial system to achieve such linguistic fluency, it must develop similar capabilities to predict and operate in conceptual spaces beyond its training data. Therefore, the aim of completely eradicating this trait presents a paradox. A model engineered for zero risk of hallucination would require disabling the inferential capabilities that allow it to handle the ambiguity and context inherent in natural language. Such a system would cease to perform *Natural Language Processing (NLP)* in its truest sense, devolving instead into a system of literal information retrieval that can only reproduce explicitly stored data.

4. Hallucination and approximation: the foundation of generative capabilities

This capability to hallucinate, far from being a flaw, is the essential mechanism that elevates LLMs beyond mere databases, giving them the power to generalize, reason creatively, and solve novel problems.

4.1. Beyond the database: Generalization through extrapolation

An LLM is a generative model, not just a data retrieval system. A traditional database stores individual records and fails when presented with a query that requires information beyond its stored knowledge. An LLM, through extrapolation, generates a rational *best guess*. This ability to go beyond the definite boundaries of its training data is precisely what we call generalization. Without these differences, an LLM would just be a lossy, inefficient database, only capable of rephrasing what it has already seen [1]. Table 2 provides a direct comparison.

4.2. The role of approximation in creative inference

The process of traversing to new vectors in the latent space is a form of algorithmic creativity. Human creativity often involves parallel reasoning and concept integration. The vector arithmetic that an LLM performs is an operational equivalent of this process. This creative potential comes directly from the same inferential mechanism that leads to factual errors. When this exploration aligns with a creative goal, we label it *inspiration*, and when it conflicts with a factual requirement, we label it *hallucination*.

4.3. The generalization–hallucination trade-off

This direct link implies a fundamental trade-off. It may be impossible to eliminate hallucination at an architectural level without impairing the model's most crucial capabilities [23]. Constraining a model to operate only within *safe*, clearly defined regions of its latent space would destroy its ability to reason analogically and handle novel prompts, reducing it to a simple retrieval system.

5. Analysis of self-correction limitations

This section presents a theoretical argument relevant to the entire class of transformer-based auto-regressive models. The analysis focuses on a fundamental limitation emerging from their deterministic architecture: the incapacity to reliably correct itself.

5.1. The Self-Correction Blind Spot

The deterministic nature of the transformer with temperature set to 0 renders the model unable of reliably fact-checking its own outputs. Research has identified a *Self-Correction Blind Spot* [20]. To analyze this accurately, it is necessary to distinguish between two distinct cognitive tasks: *Error Recognition* and *Error Correction*, and understand how complexity constrains them.

1. *Recognition vs. Correction:* The blind spot reflects a separation of capabilities. An LLM can often identify that an output is wrong ($P(\text{Error}|\text{Context}) \rightarrow 1$) but cannot reliably generate the correct value. Its review step is constrained by the same inference pathways that generated the error, so without external grounding it frequently "hallucinates a correction" instead of converging on truth.
2. *The Complexity Constraint (Why Cross-Validation Fails):* In high-complexity tasks — such as the IMO problem — self-consistency and similar cross-validation methods deteriorate. Unlike fact retrieval, where redundancy enables consensus, complex reasoning pivot on a single fragile inference path. Identical models therefore share the same deterministic bias and "collude" on the same error, rendering internal cross-validation ineffective.
3. *Variance Across Architectures:* This limitation endures across model scales. Larger models reduce initial hallucinations through denser latent spaces, but the ratio of successful self-corrections does not scale proportionally. They often generate more fluent justificatory narratives for their mistakes rather than correcting them.

Table 3
Performance metrics for initial solution attempt (IMO problem 6).

Model Name	Latency (s)	Tokens Out
Gemini Flash Latest	14.1	2804
Gemini 2.5 Flash	25.6	3777
Gemini 2.0 Flash	10.8	1555

5.2. Case study: Probing extrapolation and self-correction

To ground the theoretical model of hallucination in verifiable observation, a qualitative experiment was conducted using Google's AI Studio. The objective was to test the model's behavior on challenging problems. For instance, we selected Problem 6 from the 1988 International Mathematics Olympiad, forcing it to operate in an extrapolative, rather than retrieval-based, mode.

Problem

Let a and b be positive integers ($\mathbb{Z}^+$) such that the expression $ab+1$ divides a^2+b^2. Prove that the value of the expression

$$\frac{a^2+b^2}{ab+1} \tag{4}$$

is a perfect square.

5.2.1. Methodology

We evaluated Gemini Flash Latest, 2.5 Flash, and 2.0 Flash using the 1988 IMO Problem 6 to induce extrapolative reasoning beyond training data. Two identical instances were configured for strict determinism ($T = 0.0$, Top-P 1.0) with all auxiliary tools and chain-of-thought functionality deactivated [19]. Outputs were subsequently fed back to the same instance for self-verification. Complete experiment along with detailed methodology is accessible at https://github.com/Sarvagya-10/hallucination.

5.2.2. Observations and analysis

Performance and output metrics were recorded to quantify the generation process, as summarized in Table 3.

- *Deterministic Output:* As predicted by the model's architecture, when given identical problems, both instances of a given model produced exactly the same output, comparing each token. This confirms that the model's generation process follows a deterministic probabilistic path in the absence of artificially injected randomness.
- *Generalization as Hallucination:* On the given problem, the models failed to produce the correct solution. Instead, they gave seemingly reasonable but factually incorrect steps for solving. The generated text followed the logical structure of a mathematical proof but had faulty reasoning steps. This observation supports the central thesis that when given a query from an undefined region of its latent space, the model's attempt to generalize manifests as a hallucination—a coherent *best guess* that is ultimately baseless.
- *Refining the Recognition-Correction Gap:* In a subsequent step, the output from the first instance (Instance A) was fed to the second instance (Instance B) with a prompt asking it to "verify this solution and correct any errors." Crucially, Instance B successfully *recognized* the error, stating, "the logic in step 3 does not hold." However, it failed to *correct* it, instead offering an alternative derivation that was also factually wrong. This confirms our definition of the *Self-Correction Blind Spot*: the model is not blind to the existence of inconsistency, but it lacks the ground-truth anchor required to escape the erroneous local minimum in its latent space.

This case study demonstrates that hallucination is a practical consequence of the model's core design when it is pushed beyond the boundaries of its training data, and self-correction is not a trustworthy strategy, although self-verification may still be useful.

6. Empirical and methodological constraints

This work is primarily theoretical and mechanistic rather than a statistical benchmark. The case study using the 1988 IMO Problem 6 serves as a qualitative demonstration of the Self-Correction Blind Spot: models recognized flawed reasoning but failed to correct it, validating that error recognition and error rectification follow distinct probabilistic pathways. As a single high-complexity instance, it cannot generalize hallucination rates or provide population-level estimates. Future evaluation should analyze the recognition-to-correction ratio across model scales and task types to quantify how persistent this blind spot remains.

7. Contemporary practices for mitigation, evaluation, and application

Recognizing hallucination as an inherent result of high-dimensional extrapolation, this section examines current mitigation techniques, evaluation methodologies, and application contexts.

7.1. Mitigation strategies: grounding and structural interventions

Because hallucination stems from the same extrapolative mechanisms that enable generalization, it cannot be architecturally removed without reducing the model's reasoning ability; thus, mitigation depends on grounding and inference-level constraints. Retrieval-based systems such as RAG [23] and tool-use approaches like Toolformer [24] stablize generation through external knowledge and API-driven deterministic computation. Inference-time interventions such as Contrastive Decoding [25] adjust token probabilities to attenuate weakly supported continuations while preserving creative capacity. Weight-editing methods like ROME [26] correct specific factual associations but may destabilize adjacent semantic regions. Cross-domain techniques from Vision–Language Models further elucidate alignment-based regulation: Magic ELF improves associative consistency [10], TTST [11] filters noisy tokens, and multimodal attention interventions like CAI and CLAIM [27] constrain cross-modal drift. Collectively, these strategies show that hallucination is best managed through attention control and external grounding, rather than eliminating extrapolation.

7.2. Evaluation methodologies

Evaluating hallucination requires metrics that characterize semantic correctness and attribution fidelity, since traditional measures like ROUGE and BLEU [8] miss factual inversions and unsupported claims arising from the extrapolative mechanisms described earlier. Benchmarks such as TruthfulQA, HaluBench, FactKB, and SummEdits [1,7,28] deliver structured assessments of factualness and faithfulness, while intrinsic indicators like Perplexity Drift flag uncertainty spikes in sparsely supported latent regions where hallucinations are most likely. Ultimately, as shown in [1,16], human evaluation remains essential in high-stakes contexts, as expert judgment is uniquely able to differentiate productive abstraction from harmful error—reinforcing that hallucination must be managed, not eliminated.

7.3. Application contexts and task-tolerance profiles

Hallucination displays varying utility depending on domain tolerance. In high-tolerance settings — creative ideation, scientific hypothesis generation, algorithmic fuzzing, and exploratory design — it acts as productive stochasticity, enabling LLMs to traverse sparse latent regions and formulate novel structures or edge-case scenarios. Conversely, fields such as law or medicine demand strict factual grounding, where hallucination poses clear risk [2]. In such environments, LLMs should operate within a "Model-as-Analyst" framework that emphasizes verifiability while marking extrapolative reasoning for human review.

8. Conclusion and future scope

This work presents a mechanistic reinterpretation of hallucination as an intrinsic outcome of transformer-based generalization rather than a removable flaw. The novelty lies in formally unifying creativity, extrapolation, and hallucination through the Generalization–Hallucination Trade-off and in identifying the Self-Correction Blind Spot as an architectural limitation of deterministic auto-regressive models. Deterministic evaluations on Gemini models under zero-temperature settings show that hallucination is a reproducible artifact of latent-space extrapolation and that models can recognize but not reliably resolve erroneous reasoning without external grounding. While these findings provide qualitative evidence linking extrapolation and hallucination in high-complexity reasoning tasks, the study is constrained by controlled case-based evaluation and does not yield large-scale statistical generalization across models or domains.

In the near future, it is planned to build upon this framework by developing *Controlled Hallucination architectures* that manage the creativity–factuality balance as a runtime parameter. Future work will explore hybrid gating mechanisms that route prompts between extrapolative and evidence-grounded routes using Retrieval-Augmented Generation, extending the current analysis beyond qualitative demonstrations. Additionally, systematic testing across model scales and task categories will be conducted to assess the persistence of the *Self-Correction Blind Spot* and to quantify the proposed *Creative Dividend* on abstract reasoning benchmarks such as ARC.

CRediT authorship contribution statement

Sarvagya Singh: Writing – original draft, Resources, Methodology, Data curation, Conceptualization. **Rahul Saha:** Writing – review & editing, Supervision, Project administration, Methodology, Data curation, Conceptualization. **Gulshan Kumar:** Resources, Methodology, Data curation, Conceptualization. **Anand Nayyar:** Writing – review & editing, Visualization, Validation. **Tae-Kyung Kim:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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